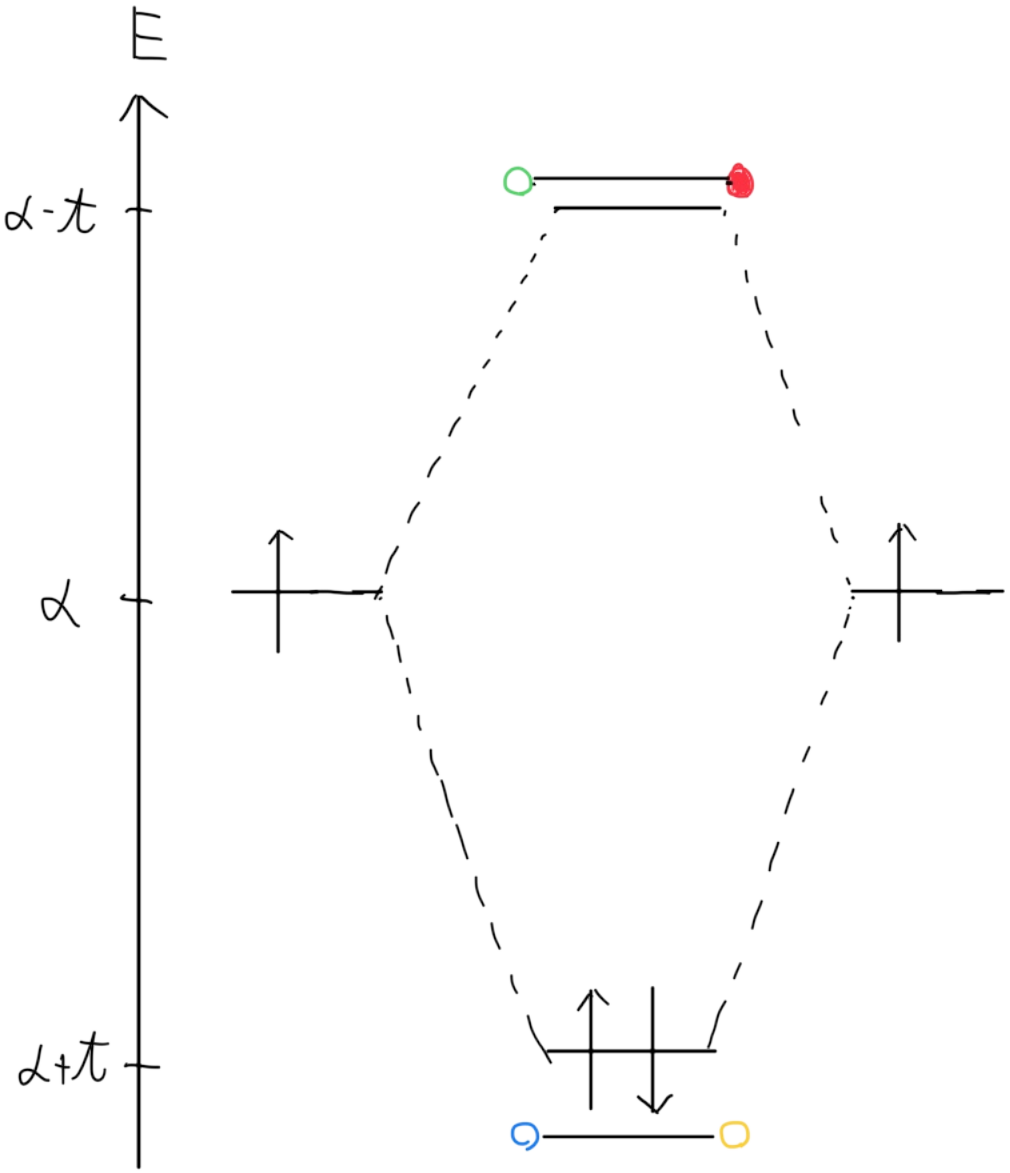

$$\text{H} \longrightarrow H_2 \longleftarrow \text{H}$$

$$\chi_A \longrightarrow c_A\chi_A + c_B\chi_B \longleftarrow \chi_B$$

$$\mathbb{C}_1 = A\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbb{C}_2 = A\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$